



"Your Partner in Innovative Refrigeration and Advanced Testing Solutions."

THERMAL SHOCK TEST CHAMBER



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WE ARE PLEASED TO OFFER THE VERTICAL THERMAL SHOCK TEST CHAMBER, AS SHOWN IN THE IMAGE, FOR THE VERIFICATION OF THE MODEL



STAND MODELS

Model			TS-015L-B2	TS-050L-B2	TS-80L-B2	TS-108L-B2	TS-216L-B2	TS-252L-B2	TS-392L-B2
Test zone size WxDxH (cm)			25x25x25	36x35x40	50x40x40	60x45x40	60x60x60	60x60x70	70x80x70
Exterior chamber size WxDxH (cm)			120 X 220 X 210	129 X 232 X 232	143 X 237 X 234	153 X 237 X 244	153 X 257 X 274	153 X 267 X 274	163 X 337 X 301
Hot zone temp. range			+50°C ~ +200°C / ± 1°C						
Cold zone temp. range			- 70°C ~ + 50°C / ± 1°C						
Performance	Temp. fluctuation		Temp. Fluctuations ±1°C at steady state in the empty basket.						
	Temp. recovery time		15-25 minutes. (Temperature range 120°C to -40°C)						
	Heating	°C	200°C						
	time	Ramp rate	1°C, 3°C, 5°C, 10°C, 15°C /minute or better (In Temperature Range: +50°C to +180°C) AS PER IEC						
	Cooling	°C	-70°C,						
	time	Ramp rate	1°C, 3°C, 5°C, 10°C, 15°C /minute (In Temperature Range: +50°C to -50°C) AS PER IEC						
Material	Exterior chamber material		CRCA Powder coating based On customer RFQ						
	Interior chamber material		SS 304 Stainless steel						
	Insulation		125 / 150 - Glass wool						
	Wind circulation		Cooling fan						
	Heating system		Stainless Steel High-speed heater						
System	Compressor / Cooling system / Condenser coil		Cascade type Air / Water cooled built-in refrigeration system, Bitzer/ Dorin / Danfoss / Bock Make Compressor, copper tube with Aluminum Fin type Evaporator or As per Customer RFQ, Air cooling condenser						
	Control system		SEMIENS PLC WITH 7" HMI WEINTEK						

TECHNICAL SPECIFICATIONS FOR THE VERTICAL / HORIZONTAL THERMAL SHOCK TEST CHAMBER

Model: TS-080 L-B2 / CZ 50 to -70 / HZ 70 to 200 / XX / A / C5 / H5 / PLC

Basket Capacity: 80 liters

Basket Dimensions (W x D x H):

50 cm x 40 cm x 40 cm

Zone Configuration:

B2: Two Zones (Hot and Cold)

Temperature Range:

CZ: +50°C to -70°C

HZ: +70°C to 200°C

Humidity Provision:

RH/XX : (No humidity provision in this model)

Cooling Type:

A: Air Cooled (Water Cooled version also available as W)

Ramp Rates:

C5: Cooling rate of 5°C/min (Empty chamber) as per IEC 60068-3-5, averaged between 50°C and -50°C.

H5: Heating rate of 5°C/min (Empty chamber) as per IEC 60068-3-5, averaged between 70°C and 180°C.

Control System:

Siemens-based PLC with profile-based control and a 7-inch HMI (Weintek / Proface).

MATERIAL SPECIFICATIONS FOR TS___ L-B2 MODEL

Exterior Chamber Material:

CRCA (Cold Rolled Close Annealed) Sheet with Powder Coating for durability and corrosion resistance.

Insulation:

High-quality Glass Wool insulation for excellent thermal resistance and energy efficiency.

Interior Chamber Material:

SS 304 Stainless Steel, ensuring high resistance to corrosion and longevity in extreme test conditions.

Air Circulation:

Equipped with a Cooling Fan to ensure uniform air distribution maintain consistent test conditions.

SYSTEM SPECIFICATIONS FOR TS___ L-B2 MODEL

Heating System

Equipped with a high-speed Stainless Steel Heater, providing rapid and efficient heating

Cooling System

Includes a **BITZER** or **Dorin** compressor, with a fin-type evaporator for effective cooling.

Air - cooled condensers ensure flexible cooling efficiency depending on the setup.



Control System

Controlled by a Siemens PLC with a 7-inch Weintek HMI touchscreen for easy operation, or available with PC-based control systems for advanced configuration without HMI.

Accessories

Observation Window:

A 300 mm W x 300 mm H Hot Zone Vacuum Glass observation window allows safe monitoring during tests.

Basket Port:

Includes an in-basket cable port (25 mm diameter) for easy wiring access.

Additional Ports:

Features 75 mm diameter ports for both Hot and Cold zones.

Lighting & Racks:

Test chamber includes internal lighting for better visibility and comes with 1 rack.



SAFETY FEATURES FOR COMPRESSORS AND THERMAL SHOCK TEST CHAMBERS

High-Pressure Trip

Function: Triggers an automatic shutdown of the compressor when pressure exceeds the preset limit.

Protection Mechanism: Monitored using pressure sensors integrated with the PLC system.

Low-Pressure Trip

Function: Activates when the pressure falls below the designated threshold, ensuring safe operation of the compressor.

Protection Mechanism: Managed by pressure transducers connected to the PLC for real-time monitoring.

Compressor Winding Temperature Protection

High-Temperature Trip:

Function: Prevents damage to the compressor winding by shutting down the unit if the temperature exceeds safe levels.

Protection Mechanism: Utilizes thermistors for accurate temperature measurement connected to the PLC system.

Ambient Temperature Protection

Ambient High-Temperature Trip:

Function: Automatically shuts down the system if the ambient temperature exceeds a set limit, protecting sensitive components.

Protection Mechanism: Monitored using an RTD (Resistance Temperature Detector) connected to the PLC.

Overload Protection for High-Speed (HS) and Low-Speed (LS) Compressors

Function: Protects motors from overheating by shutting down the system in case of overload conditions.

Protection Mechanism: Integrated with the PLC to monitor motor currents and operational parameters.

Single Phase Prevention Trip

Function: Automatically shuts down the system to prevent damage if the compressor operates in a single-phase condition.

Protection Mechanism: Monitored through phase detection circuits connected to the PLC.

Temperature Monitoring for Hot and Cold Zones

High-Temperature Trip for Hot Zone:

Function: Protects against excessive temperatures in designated hot zones, ensuring safe operation.

Low-Temperature Trip for Cold Zone:

Prevent operation dangerously low temperatures cold zones to protect equipment integrity.

Heater Short Circuit Trip

Function: Protects against electrical failures, shutting down the system in the event of a short circuit in the heating elements.

Protection: Integrated circuit protection systems linked to the PLC for monitoring.

SUBSYSTEM DETAILS

Expansion Valve, Dryers, Oil Separators, Pressure Transmitter, and Vibration Isolators:

These key components will be sourced from **reputable international manufacturers** such as **Danfoss, Emerson, Bitzer, Castel, Bummer, and Dryall** ensuring reliability and top performance.

Evaporator and Heater Material:

The **evaporator** is designed with **copper tubes** and **aluminum fins** for efficient heat exchange and long-lasting durability.

Heating elements are constructed using **stainless steel** or **Inconel sheathing** offering excellent corrosion resistance. The system includes **redundant over-temperature protection** for added safety.

Refrigerant:

The chamber's cooling system utilizes a **water-cooled cascaded refrigeration system**, employing **R404A** for the high stage and **R23** for the low stage refrigerants.

The system can be operated in **auto mode or manual mode** using the advanced **PLC system**, offering continuously variable power adjustments for precise control over the temperature environment.

CONTROL SYSTEM

The chamber control system features:

Dual PT100 A-Class temperature sensors for accurate and reliable temperature readings.

Siemens PLC S7-1200 integrated with a **7" Weintek HMI**, providing an intuitive user interface for easy control.

Connectivity options include **LAN** and **USB ports**, along with an **independent temperature controller** to ensure precise temperature regulation.

An **optional remote operation feature** with the **SESS Climatic Test Viewer PC software** is available, allowing users to monitor and control the chamber from a remote location.

MATERIAL OF CONSTRUCTION

Interior Material

The inner chamber will be constructed from **AISI grade SS304 stainless steel**, known for its **corrosion resistance** and durability. The **mirror-finish sheets** have a thickness of **1.5 mm** (some parts may be **1.2 mm**), and all joints are **TIG-welded** for added strength and a smooth finish. The **inner wall** has minimal welded joints to ensure a clean, sleek surface.

Exterior Material

The outer wall of the chamber is made of **CRCA steel** (Cold Rolled Close Annealed) and is **powder-coated** for enhanced durability and aesthetic finish.

Insulation

The test space will be **well-insulated** to ensure thermal efficiency, and all necessary measures will be taken to prevent **moisture condensation** on the external surface of the chamber. The insulation is made of **glass wool** with a thickness of **150 mm**, providing excellent thermal stability.

SHELF FOR LOADING TEST ARTICLES

A removable and **height-adjustable stainless steel shelf** with a **load rating of 20 kg** will be provided. The shelf allows flexible positioning for different test requirements.

Chamber Inner Viewing Lamp:

A **20-watt lamp** will be installed at the **viewing window** for easy monitoring of the chamber's interior. The lamp is operated via the **HMI control** for convenience.

Door Gasket:

The door will be sealed with a **double-layer silicon gasket** to ensure airtight sealing. The design includes one layer within the door and an additional outer layer for maximum **insulation** and **temperature control**.

Hinges & Latches:

The main door will be equipped with **heavy-duty hinges**, hinged on the left side, with secure locking mechanisms. The chamber will also feature **removable side access panels** (not hinged), allowing easy maintenance and access. Each panel will have a **round insert panel lock** for added security.

Mounting:

The chamber will be a **vertical floor-mount type**, equipped with **four heavy-duty, lockable caster wheels** capable of carrying **800 kg per wheel**. This ensures easy mobility, while **vibration isolation** mechanisms will be provided to stabilize the chamber during operation. Provisions will be made to anchor the chamber, preventing movement during testing.



INDEPENDENT SAFETY TEMPERATURE CONTROLLER

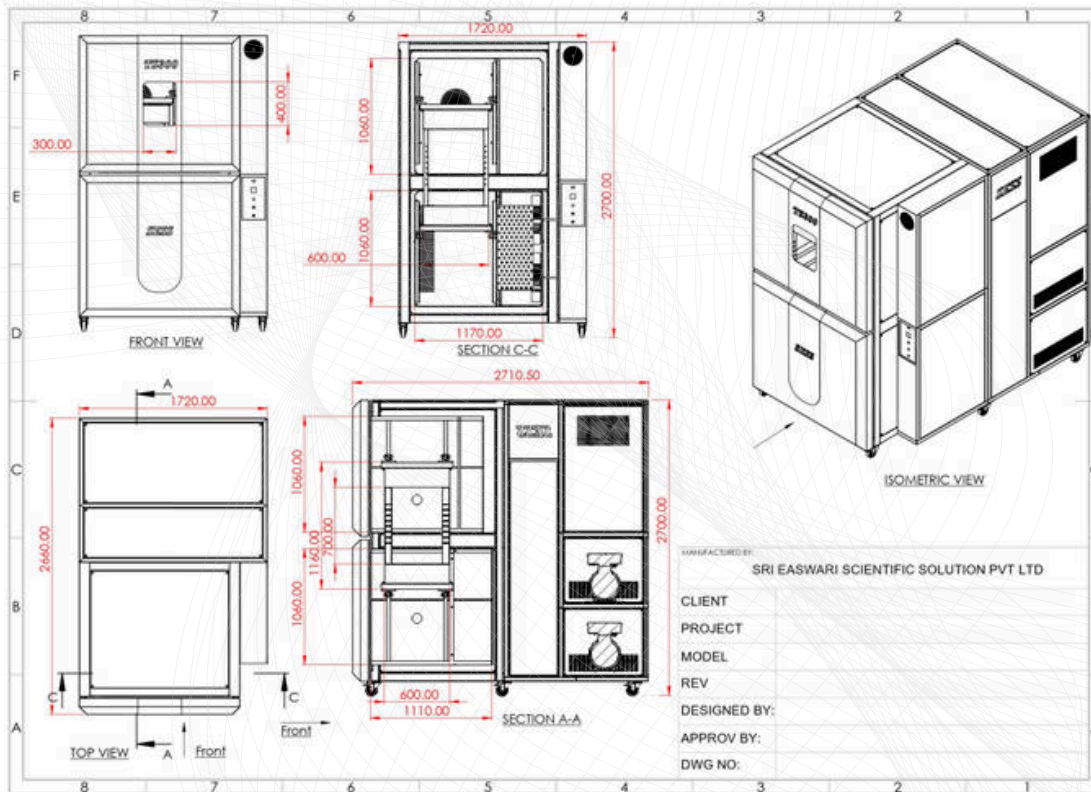
An **independent controller** will be included to limit the **maximum temperature** inside the chamber. If the set temperature exceeds the safe limit, the chamber will **automatically shut down** to ensure safety.

Power Supply: Operates on **AC 3-phase 415V $\pm$ 5%**, 50Hz with neutral and ground (N+G), ensuring stable power input.

Customization:

Non-standard or special requirements:

Customization options are available upon request to meet specific testing needs.



The GA (General Arrangement) for a few TS-252 L models is provided for reference only (Basket Inner size is 60 x 60 x 70 cm). Once a

Purchase Order (PO) is received, WE will provide the GA for the requested model.

ELECTRICAL PANEL SPECIFICATIONS

Design and Components:

Switchgear Items: The electrical panel is designed to house all necessary switchgear components, including:

- Separate MCBs (Miniature Circuit Breakers)
- Contactors
- Overload Relays (OLR) / Motor Protection Circuit Breakers (MPCBs) for fan motors and refrigeration compressors
- MCBs, contactors, and separate Solid State Relays (SSRs) for each phase of air heaters.

Sound Pressure Level:

- **Noise Level:** Less than 80 dB (A) at a distance of 1 meter from the front door.

Safety Features:

Fan/Blower Failure Alarm:

An alarm is triggered in the event of fan/blower failure, leading to automatic chamber shutdown and clear display indications on the control panel.

Protection Mechanisms:

- Phase reversal protection
- Single phasing protection
- Overcurrent and short circuit protection

Electrical Connection:

Configuration:

3 Phase + Neutral + Earth

Voltage: 415V AC, 50Hz

Control System:

HMI Integration: A soft-touch-based control system has been implemented within the Human-Machine Interface (HMI), enabling full automation. This system allows for:

- Seamless activation during profile creation via an event-based option

Fault Diagnosis Capabilities:

The control system includes advanced Fault Diagnosis features:

- A diagnostic display for tracking parameters and input/output statuses
- A Help menu for user assistance
- Alarm history for review and analysis

DESCRIPTION - CHAMBER CONSTRUCTION

The chamber is meticulously crafted with a double-walled, insulated design using argon Tig/arc welded stainless steel sheet (304) measuring 1.5/1.2 mm in thickness for the interior, while the outer walls are constructed using CRCA, as specified in the customer's tender RFQ sheet, with a thickness of 1.5 mm. The outer body is manufactured using CNC technology, ensuring precise production and featuring seamlessly smooth joints without any protruding fixtures.

The chamber's door is coated with an epoxy-based powder in RAL 9001 cream white, while the main chamber body is finished with RAL 7024 graphite grey powder coating. The chamber is ingeniously designed in a vertical trolley mounting configuration, featuring a fully frontal opening double-walled insulated door. This door is thoughtfully equipped with an exclusively sourced high-quality silicone rubber gasket for optimal sealing. To ensure both security and insulation, the door is fitted with robust hinges and a toggle-type locking mechanism.

This meticulous attention to construction details not only underscores the chamber's durability and functionality but also enhances its aesthetic appeal.



IN OUR CLIMATIC TEST CHAMBER'S AIR CIRCULATION SYSTEM, WE INCORPORATE THE FOLLOWING COMPONENTS AND FEATURES

Fan/Blower: The chamber is equipped with a fan or blower featuring a continuous-duty rated motor. This motor efficiently circulates air within the chamber.

Conditioning Space: Inside the main chamber, there's a dedicated conditioning area with baffles, housing essential components like heaters and cooling coils. These elements play a crucial role in air conditioning.



Uniform Temperature: Conditioned air is introduced into the main chamber to ensure consistent temperature distribution throughout the workspace, maintaining a stable temperature environment during tests.

Laminar Airflow: The chamber is designed for laminar airflow, achieved using a fan motor and impeller with an appropriate capacity. This setup guarantees even distribution of temperature and humidity within the chamber, adhering to specified limits.

Motor Shaft: The motor shaft used for chamber air circulation is constructed with SS 304 without any joints. The motor is positioned outside the chamber, while an extended-length SS 304 shaft drives the fans. This design ensures smooth and reliable operation while preserving the chamber's internal environment.

Motor Winding: For high-temperature chambers, we utilize motor windings of H class, whereas for chambers operating below 130°C, F class motor windings are employed.

These features combine to create an efficient and precise air circulation system within the climatic test chamber, facilitating accurate and dependable test conditions for various experiments and applications.

INCONEL-TYPE AIR HEATERS WITH PTFE WIRE AND SILICON END CAP

Heating System: The chamber is equipped with sealed inconel tube heaters to maintain a uniform temperature. These heaters are meticulously designed to ensure even heat distribution throughout the chamber. Heat input is regulated by solid-state relays, controlled through a PLC with a PID controller programmer.

One notable advantage of inconel tube heaters is their exceptional ability to withstand high temperatures and corrosive environments. Inconel, known for its outstanding durability and heat resistance, is well-suited for prolonged use in challenging conditions.



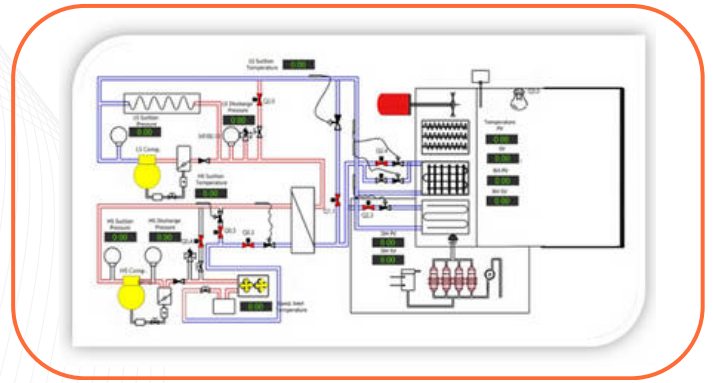
Compared to stainless steel heaters, inconel heaters exhibit superior resistance to oxidation, corrosion, and high-temperature degradation. They are engineered to deliver reliable and efficient heat transfer without direct radiation onto the test item, ensuring a uniform heating process that doesn't adversely affect the test specimen.

In conclusion, the use of sealed inconel tube heaters is highly recommended for maintaining precise and uniform chamber temperatures while ensuring the long-term performance and durability of the heating system.

OUR CHAMBER'S REFRIGERATION SYSTEM IS DESIGNED FOR OPTIMAL PERFORMANCE

- Single-stage compressor for temperatures up to -40°C, and a double-stage compressor for temperatures below -70°C. We select our compressors from reputable brands like Bitzer, Dorin, Bock, Danfoss, SECOP, or Embraco.
- Environmentally friendly refrigerants are used, with R404A for single-compressor systems and R404A/R23 for cascade systems.
- Safety features include HP/LP cut-off switches, pressure transmitters for continuous monitoring, and suction line temperature sensors for enhanced safety.

- The system can have a water or air-cooled condenser based on the model. Temperature monitoring and safety limits are integrated into our PLC system for effective control and protection.
- The compressor and condenser unit are positioned below the chamber for optimal performance.
- For condenser cooling, our PLC system monitors the air or water inlet temperature. If it exceeds 26°C (water) or 40°C (air), a safety alarm triggers, and chamber operation stops until the temperature returns to normal, which can be adjusted through HMI and PC software. This setup ensures efficient cooling and temperature control for our thermal cyclic test chamber.



Our test chamber's instrumentation includes a Siemens S7 1200 PLC and a custom-designed HMI by Proface, Weintek, or SESS. The HMI boasts a 7" touchscreen with segment 100 functionality, allowing for programming and profiling of up to 10 cycles with a maximum of 999 repetitions, catering to customer RFQ requests.

Furthermore, we offer an alternative model equipped with a PC-based control system. Our PLC communicates seamlessly with PC LabVIEW software through Ethernet TCP/IP. This LabVIEW-based software, known as SESS Climatic Test Viewer, operates without the need for a license, requiring only a free runtime engine. It also offers unlimited storage capabilities for programming and control log files. To ensure precise temperature measurement and control, we employ PT100 temperature sensors of class A type. The HMI simplifies programming and control of the test chamber.

Additionally, chamber operation can be carried out via PC software. For PC-based control, the following requirements apply (Note that if you opt for the HMI model, the PC is optional and will be provided upon your request):

- HP or Dell Desktop Computer
- Processor: Intel Core i3 or higher
- Memory (RAM): 8 GB
- Internal HDD Capacity: 512 GB
- Display Monitor: 21.5" FHD (Full HD)
- At least one dedicated Ethernet port for chamber connection and an additional Wi-Fi Ethernet port for network connectivity.

PC Software Future

- Test parameter Data logging
- Make a multiple Profile
- Refrigeration Performance monitor window
- DI/DO'S status window monitoring
- Control system monitor screen to identify the working condition status of solenoid valves, pressure status and temperature display .

This combination of the PLC control system, custom HMI, and LabVIEW-based PC software offers a versatile and comprehensive solution for programming, controlling, and monitoring test chamber operations. It ensures flexibility and ease of use, facilitating efficient and precise temperature testing across various applications.

Control Panel & Wiring: A separate control panel will be provided, attached to the main chamber, which will house the programmer, on/off switches, fuses, contactors, indicating lamps, etc. Channel type wiring will be implemented using suitable copper wires with appropriate current ratings, along with marking ferrules, crimped dowel terminals, elmex connectors, and so on.

The electrical and instrumentation panel will be meticulously wired in compliance with international standards to ensure safety and reliability. Each electrical cable will be tagged with ID numbers, and the corresponding details will be clearly documented in the maintenance manual. Comprehensive circuit diagrams and relevant information will also be included.

To facilitate accurate and efficient design, we utilize SolidWorks 3D software for developing our drawings. This advanced software enables us to create detailed structural and electrical drawings that adhere to industry standards and specifications. By utilizing SolidWorks, we can ensure precise and comprehensive documentation of the electrical components and wiring layout of the Thermal cyclic test chamber.

Furthermore, we take pride in mentioning that our standard test chamber has been certified by TUV and carries the CE mark. This certification validates that our Stand test chamber only complies with the essential health, safety, and environmental protection requirements of the European Union. It serves as a testament to our commitment to providing high-quality, safe, and reliable equipment.

Our dedication to utilizing modern software tools and obtaining relevant certifications reflects our unwavering commitment to delivering superior products that meet the highest industry standards and exceed customer expectations.

Solid state relays for control for dry heater.

IN OUR CHAMBER, THE PLC (PROGRAMMABLE LOGIC CONTROLLER) WILL EFFICIENTLY CONTROL AND MONITOR THE FOLLOWING ASPECTS:

Temperature sensors: The PLC will handle the set point temperature and continuously monitor the attained temperature values to ensure precise temperature control.

Status of compressor: The PLC will monitor the operating status of the compressor, including start/stop commands and fault detection, to ensure proper functioning of the refrigeration system.

Pressure sensors: The PLC will gather data from pressure sensors to monitor the pressure levels within the system and trigger appropriate actions based on predefined thresholds.

Valves: The PLC will control various valves in the system, such as expansion valves, solenoid valves, and safety valves, to regulate the flow of refrigerant and other fluids.

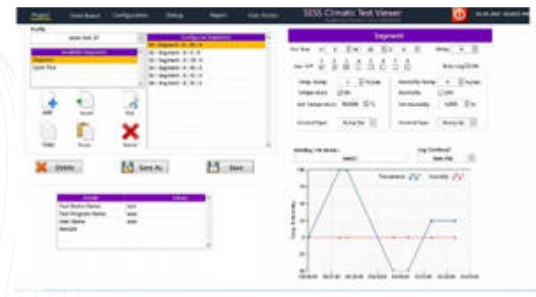
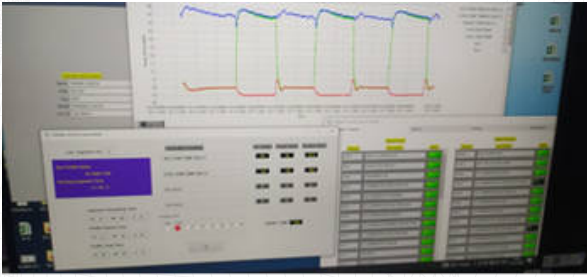
Heater: The PLC will control the operation of the heater, ensuring accurate and consistent heating as per the desired temperature profile.

Circulation Fan / Blower: The PLC will regulate the speed and operation of the circulation fan or blower to maintain proper air circulation and temperature uniformity within the chamber.

Over current / over temperature protection: The PLC will incorporate protective measures, such as overcurrent and over temperature protection, to safeguard all the above components from potential damage or malfunctions.

By centralizing the control of these critical components, the PLC ensures efficient and reliable operation of the thermal cyclic test chamber while maintaining safety and accuracy in temperature control.

CONTROL DISPLAY MAIN SCREEN



PC SOFTWARE IOS DETAIL AND ALARM LOG SCREEN

Safety protection: (**Note:** Based on model, this may vary.)

- Over temperature safety cut off thermostat with audiovisual alarm.
- High temperature Alarm
- Low temperature Alarm
- HP/LP cut off switch for compressor./ Comp thermal Protection Alarm.
- Overload protector for motor.
- MCB for heaters.
- Refrigerant low Alarm.
- Single Phase Preventers / Equipment fault / control Off .
- High temperature alarm for Ambient/Condenser inlet
- Air Heater protection etc....



CERTIFICATION

ISO 9001-2015



TRADEMARK



MSME





THANK YOU!

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